

Averaged ionizing emissivity of stellar photons and cosmic-ray electrons, counting all ionizations down the cascade Hydrogen and helium branches

Response note — `State of the art - Reionization/ionizing_emissivity/`

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Abstract

We compute, separately for hydrogen and for helium, three quantities as functions of the energy of the primary particle: the number of ionizations produced per primary particle counting every ionization down the cascade; the differential ionizing emissivity per comoving volume; and the ionization rate per target atom. Two primary populations are treated, stellar Lyman-continuum photons and cosmic-ray electrons, and two independent cascade engines are run against each other throughout. Every numerical input is taken from a named paper; the only quantities computed rather than quoted are two integrals (an IMF integral and a virial-theorem binding energy) that are performed from scratch here and then checked against their published values.

The result is that cosmic-ray electrons deliver $(3.8\text{--}7.5) \times 10^{-5}$ of the hydrogen-ionizing emissivity of stellar photons, and $(2.0\text{--}4.4) \times 10^{-5}$ of the helium-ionizing emissivity, essentially independently of redshift over $6 \leq z \leq 15$ and of the residual ionized fraction over $10^{-4} \leq x_e \leq 10^{-1}$. Including the full range spanned by the two published cosmic-ray injection prescriptions and their efficiency uncertainties, the hydrogen share is $5.8 \times 10^{-6}\text{--}4.7 \times 10^{-4}$ and the helium share $3.1 \times 10^{-6}\text{--}2.7 \times 10^{-4}$.

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1 Assumptions

Every choice below was put to the user before any number was computed; none was made unilaterally. Where an assumption is known to be internally inconsistent it is marked **flagged** and carried explicitly into the results.

1.1 Cosmology and composition

Flat Λ CDM with [Planck Collaboration \(2020\)](#) Table 1 (TT,TE,EE+lowE+lensing): $\Omega_b h^2 = 0.02237$, $h = 0.6736$, $Y_p = 0.2454$. Hence, computed in `emis_common.py`,

$$\langle n_H \rangle_{\text{com}} = 1.8946 \times 10^{-7} \text{ cm}^{-3}, \quad \frac{n_{\text{He}}}{n_H} = \frac{Y_p}{4X_p} = 0.08130. \quad (1)$$

1.2 Ionization thresholds and cross-sections

Thresholds from NIST: 13.598, 24.587, 54.418 eV for H I, He I, He II. Photoionization cross-sections from [Verner et al. \(1996\)](#), Eq. (1) and Table 1, selected by the user over the Osterbrock & Ferland fits:

$$\sigma(E) = \sigma_0 F(y), \quad y = \sqrt{(E/E_0 - y_0)^2 + y_1^2}, \quad F(y) = [(y-1)^2 + y_w^2] y^{0.5P-5.5} (1 + \sqrt{y/y_a})^{-P}. \quad (2)$$

Validity: the fits are constructed to $E_{\text{max}} = 5.0 \times 10^4$ eV for all three ions. Above 50 keV Eq. (2) is still used, which reproduces the correct *non-relativistic* asymptote $\sigma \propto E^{-3.5}$ by construction; this is marked on every panel (a). Above ~ 511 keV Compton scattering dominates the total opacity in any case.

1.3 The helium ionization state

By explicit instruction, helium is taken FULLY NEUTRAL, $n_{\text{HeI}} = n_{\text{He}}$, at every x_e . This is an upper bound on the helium target density and is internally inconsistent with $x_e > 0$. It is stated on every helium panel and in every helium table. The alternative standard assumption, $x_{\text{HeII}} = x_{\text{HII}}$, would reduce the helium emissivity by a factor $(1 - x_e)$, i.e. by at most 10 per cent over the x_e range considered, so this choice does not change any conclusion drawn here.

1.4 The two cascade engines

Engine A — this project’s own transport cascade. Two distinct yields are kept separate because they answer different questions:

- **A-csda:** in-situ *collisional* ionizations only, from the continuous-slowng-down solver `Notebooks/redo/stage_csda.py`, whose loss physics is inverse Compton and bremsstrahlung ([Blumenthal & Gould, 1970](#)), Coulomb losses ([Gould, 1972](#)), collisional excitation ([Stone & Kim, 2002](#)) and RBEB collisional ionization ([Kim et al., 2000](#)). Inverse Compton, bremsstrahlung and adiabatic losses are booked as escape. The solver’s grid is *extended here* from its default ceiling of 1 MeV to 10 TeV. Available at every x_e and z .
- **A-transport:** the above *plus* the ionizations produced by the photons the electron itself radiates and that are re-absorbed inside the reionization window (column $Y + Y_p^{(e)}$ of `redo_cascade_table.npz`). Available at $x_e = 10^{-4}$ only.

Engine B — the published prescriptions. Each is used only inside its own stated validity range; outside it the code returns NaN and the curves simply stop.

- [Shull & van Steenberg \(1985\)](#), Table 2 — *the only source in the literature that resolves the H I/He I split in closed form*. Form 1 $y = C[1 - (1 - x^a)^b]$ for heat, Form 2 $y = C(1 - x^a)^b$ for ionization and excitation, accurate to 1–2 per cent over $10^{-4} < x < 1$, fitted to the $E_0 > 100$ eV limiting curves with the Monte Carlo run to 3 keV.
- [Valdés & Ferrara \(2008\)](#), Eqs. (4)–(7), accurate to 3.5 per cent; $E_{\text{in}} = 3$ and 10 keV; *total* ionization only.
- [Valdés et al. \(2010\)](#), Appendix A, Tables A1–A21; 1 MeV–1 TeV at $z = 10, 30, 50$ on a ten-point x_e grid; *total* ionization only. This is the only external source covering the range where inverse Compton dominates.

Three caveats carried into every engine-B result: (i) [Shull & van Steenberg \(1985\)](#) assume $n(\text{He})/n(\text{H}) = 0.1$, 23 per cent above the Planck value used here; (ii) they also assume $n(\text{He}^+)/n(\text{He}_{\text{tot}}) = n(\text{H}^+)/n(\text{H}_{\text{tot}})$, which contradicts the fully-neutral helium adopted above and cannot be undone from the published fits; (iii) they state that “excitations or ionizations of He II were negligibly small”, so engine B provides no He II channel.

1.5 Source normalisations

Photons. Two, both measured.

- [Gaikwad et al. \(2023\)](#), Table 3, $z = 6.00$, from the Ly α forest: $\dot{n}_{\text{ion}} = 0.701 (+0.357 / - 0.191) \times 10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$, $\Gamma_{\text{H I}} = 0.145 (+0.157 / - 0.087) \times 10^{-12} \text{ s}^{-1}$, $\alpha_s = 2.0 \pm 0.6$.
- [Giovinazzo et al. \(2026\)](#), Table B.1, from JWST/NIRSpec: $\log_{10}(\dot{n}_{\text{ion}}/\text{s}^{-1} \text{ Mpc}^{-3})$ at integer z , from 51.12 ± 0.09 at $z = 5$ to $49.30 (+0.35 / - 0.29)$ at $z = 15$.

The photon *number* spectrum follows from $\epsilon_\nu \propto \nu^{-\alpha_s}$:

$$\frac{d\dot{n}}{dE} = \dot{n}_{\text{ion}} \alpha_s E_{\text{th}}^{\alpha_s} E^{-\alpha_s-1}, \quad E \geq E_{\text{th}}(\text{H I}), \quad (3)$$

normalised so that its integral above 13.598 eV returns $\dot{n}_{\text{ion}}$ exactly (verified symbolically, Sect. 3A). [Giovinazzo et al. \(2026\)](#) **quote no spectral index**, so $\alpha_s = 2.0$ is used for the shape in both cases.

Cosmic-ray electrons. Tied to the *same* star formation that makes the photons, by instruction:

1. invert [Robertson et al. \(2015\)](#) Eq. (1), $\dot{n}_{\text{ion}} = f_{\text{esc}} \xi_{\text{ion}} \rho_{\text{SFR}}$, with their fiducial $f_{\text{esc}} = 0.2$ and $\log_{10} \xi_{\text{ion}} = 53.14 [\text{s}^{-1} \text{ per } M_\odot \text{ yr}^{-1}]$;
2. convert to a supernova energy rate with $E_{\text{SN}} = 10^{51} \text{ erg}$ ([Woosley & Weaver, 1995](#)) and the stellar mass per core-collapse supernova *computed here* by integrating the [Chabrier \(2003\)](#) disc IMF over 8–100 $M_\odot$ progenitors: $M_*/N_{\text{CCSN}} = 84.75 M_\odot$;
3. apply the efficiencies collected by [Gessey-Jones et al. \(2023\)](#): cosmic rays carry $\eta_{\text{cr}} = 10$ –50 per cent of the shock kinetic energy ([Drury et al., 1989](#); [Berezinskii et al., 1990](#); [Caprioli & Spitkovsky, 2014](#)), with “per cent level” in electrons, taken as $f_e = 0.01$ –0.02;
4. distribute that power over two published injection spectra, which together define the band: [Tueros et al. \(2014\)](#), $dN/dE \propto E^{-2.2}$ ([Drury 1983 DSA](#)), 1 MeV–100 TeV; and [Gessey-Jones et al. \(2023\)](#) Eq. (23), $dN/dE \propto E^{-2.0}$, 1 keV–1 PeV.

2 Derivation

2.1 Ionizations per primary particle

For an *electron* primary, engine A supplies the total ionization yield $N_{\text{tot}}(K, z, x_e)$ and the species split is derived rather than assumed. In the continuous-slowng-down approximation the number of ionizations of species i produced as the electron degrades is

$$N_i(E_0) = \int_{B_i}^{E_0} \frac{n_i \sigma_i^{\text{BEB}}(E)}{\Lambda(E)} dE, \quad (4)$$

with Λ the total energy-loss rate per unit path, common to both species. The split $f_H(E_0) = N_{\text{HI}}/(N_{\text{HI}} + N_{\text{HeI}})$ is therefore a ratio of two integrals over the same Λ , and

$$N_{\text{HI}} = f_H N_{\text{tot}}, \quad N_{\text{HeI}} = (1 - f_H) N_{\text{tot}}. \quad (5)$$

The cross-sections in Eq. (4) are Binary-Encounter-Bethe (Kim & Rudd, 1994), the same model already used for hydrogen in this project’s `igm_losses.py`:

$$\sigma(T) = \frac{S}{t + u + 1} \left[\frac{\ln t}{2} \left(1 - \frac{1}{t^2} \right) + 1 - \frac{1}{t} - \frac{\ln t}{t + 1} \right], \quad t = \frac{T}{B}, \quad u = \frac{U}{B}, \quad S = 4\pi a_0^2 N \left(\frac{R}{B} \right)^2. \quad (6)$$

B and N are from NIST and are exact. U , the mean orbital kinetic energy, is *not* taken from a table: it is derived from the virial theorem for a Coulomb-bound system, $\langle T \rangle = -E_{\text{total}}$, giving $U(\text{HI}) = 13.598$ eV and $U(\text{HeI}) = (24.587 + 54.418)/2 = 39.5025$ eV. The helium value reproduces the 39.51 eV tabulated by Kim & Rudd (1994) to 0.019 per cent, which is an independent check that the derivation is the right one.

For a *photon* primary the absorption branches between the two neutral species with the Verner cross-sections, and the photoelectron then cascades:

$$N_j^\gamma(E) = P_j(E) [1 + N_j^e(E - E_{\text{th},j})] + \sum_{k \neq j} P_k(E) N_j^e(E - E_{\text{th},k}), \quad P_k = \frac{n_k \sigma_k}{\sum_l n_l \sigma_l}. \quad (7)$$

2.2 The three plotted quantities

With $d\dot{N}/dE$ the volumetric injection spectrum of a population,

$$(a) \quad N_{\text{ion}}(E) \quad [\text{dimensionless}], \quad (8)$$

$$(b) \quad \frac{d\dot{n}_{\text{ion}}}{d \log_{10} E} = \ln(10) E \frac{d\dot{N}}{dE} N_{\text{ion}}(E) \quad [\text{s}^{-1} \text{ cMpc}^{-3} \text{ dex}^{-1}], \quad (9)$$

$$(c) \quad \frac{d\zeta}{d \log_{10} E} = \frac{1}{n_{\text{target, com}}} \frac{d\dot{n}_{\text{ion}}}{d \log_{10} E} \quad [\text{s}^{-1} \text{ per atom dex}^{-1}]. \quad (10)$$

Both $\dot{n}_{\text{ion}}$ and n_{target} are comoving, so ζ is a proper rate per atom and is redshift-invariant in that ratio.

3 Verification and cross-checks

Full output: `ionizing_emissivity/emis_verify.log`. Blocks A–E are pass/fail tests and all pass; F–J are comparisons and findings.

A — source normalisation (sympy). $\int_{E_{\text{th}}}^{\infty} \dot{n} \alpha E_{\text{th}}^\alpha E^{-\alpha-1} dE$ simplifies to $\dot{n}$ exactly, and the fraction above E_2 simplifies to $(E_2/E_{\text{th}})^{-\alpha}$ — the same identity already verified in `photon_energy_fractions.py`. Numerically the implemented spectrum integrates to $\dot{n}_{\text{ion}}$ with ratio 1.0000000000.

B — dimensions. $[E \, d\dot{N}/dE \, N_{\text{ion}}] = (\text{time} \cdot \text{volume})^{-1}$ and $[\cdot/n_{\text{target}}] = \text{time}^{-1}$, both confirmed with `sympy.physics.units`.

C — the BEB U from the virial theorem. $U(\text{He I}) = 39.5025 \text{ eV}$ derived, against 39.51 eV tabulated by [Kim & Rudd \(1994\)](#): 0.019 per cent.

D — Verner cross-sections. Threshold values $6.346, 7.435, 1.587 \times 10^{-18} \text{ cm}^2$ for H I, He I, He II, against the independent literature values $6.30, 7.42, 1.58 \times 10^{-18}$: agreement to 0.7, 0.2 and 0.5 per cent. The hydrogenic Z -scaling that must relate a one-electron ion to hydrogen, $\sigma_Z(E) = \sigma_{\text{H}}(E/Z^2)/Z^2$, is satisfied by the independent He II fit to 0.18 per cent from 60 eV to 10 keV. The logarithmic slope over 20–40 keV is $-3.47, -3.46, -3.44$ against the required -3.5 .

E — engine A closure. The CSDA identity $BY + H + X + E_{\text{esc}} = K$ is checked at every energy of all 24 runs ($4 \, x_e \times 6 \, z$): worst relative residual 1.9×10^{-3} , median 7.3×10^{-4} , largest at the low-energy end of the grid where the marching error accumulates.

K — pass/fail on the headline quantity. Blocks F–J are comparisons and cannot fail. A mutation test (§3, `CHECKS_AUTO C23`) showed that before this block a factor-1.5 error in engine A’s electron yield — the central number of the whole calculation — passed the entire suite silently. Block K now asserts engine A against [Shull & van Steenberg \(1985\)](#) over 300 eV–3 keV, where the two are independent calculations of the same quantity, requiring the ratio to lie in $[0.90, 1.10]$ for $x_e \leq 10^{-2}$. A deliberate $\times 1.08$ perturbation makes it fail, so the band retains discriminating power. $x_e = 10^{-1}$ is excluded and reported rather than absorbed into a wider tolerance.

F — engine A against engine B. Inside [Shull & van Steenberg \(1985\)](#)’s computed range and for H I at $z = 10$, A-csda/SvS85 over 300 eV–3 keV is $0.950\text{--}0.964$ ($x_e = 10^{-4}$), $0.989\text{--}1.002$ (10^{-3}), $0.956\text{--}1.019$ (10^{-2}) and $0.701\text{--}0.943$ (10^{-1}). Extending down to 100 eV the worst case degrades to 0.409, at $x_e = 10^{-1}$, where SvS85’s $E_0 > 100 \text{ eV}$ limiting fits are at their edge. Above 1 MeV, A-transport/[Valdés et al. \(2010\)](#) over the full 1 MeV–1 TeV overlap spans $0.264\text{--}2.63$ ($x_e = 10^{-4}$), $0.272\text{--}2.75$, $0.304\text{--}3.30$ and $0.425\text{--}6.32$ (10^{-1}). *The two engines therefore agree to about five per cent over 300 eV–3 keV for $x_e \leq 10^{-2}$, and to a factor $\lesssim 3$ at MeV–TeV energies*; this is the systematic uncertainty of the whole calculation and is shown as Fig. 3.

G — literature finding: energy closure of [Valdés et al. \(2010\)](#) Appendix A. The paper states its last column is “a test of energy conservation”. Summing the five deposition columns plus the CMB column gives $1.0000\text{--}1.0002$ at $E_{\text{in}} = 1 \text{ MeV}$ — which confirms the column identification used here — but only $0.77\text{--}0.94$ at $10\text{--}100 \text{ MeV}$. The missing energy is inverse-Compton photons between 10.2 eV and 10 keV, which fall in no tabulated column. This is a property of the published tables and it limits how tightly engine B constrains engine A in that decade.

H — literature finding: the CMB in [Tueros et al. \(2014\)](#). Their Sect. 2 states, verbatim, “The CMB has been considered monoenergetic with photon energy $E_{\text{CMB}} = 3.75 \times 10^{-4}(1 + z) \text{ eV}$, and a photon density $u_{\text{CMB}} = 0.05(1 + z)^3 \text{ cm}^{-3}$.” Computing the measured CMB from $T_0 = 2.7255 \text{ K}$ ([Fixsen, 2009](#)) gives, at $z = 0$, $n_\gamma = 410.73 \text{ cm}^{-3}$, $u_\gamma = 0.2606 \text{ eV cm}^{-3}$ and $\langle E \rangle = 6.344 \times 10^{-4} \text{ eV} = 2.701 kT$. Their photon density is therefore 8.2×10^3 times below n_γ and their implied energy density 1.4×10^4 times below u_γ ; read instead as an energy density in eV cm^{-3} , the shortfall is a factor 5.2. Either reading under-estimates the inverse-Compton loss rate, which is linear in u_{CMB} . This is stated as an observation about the published parameters, not as a re-derivation of their result.

J — the Tueros et al. (2014) benchmark. Their Eq. (2), $I_{E_c} = E_c \int I(E) E^{-\beta} dE / \int E^{1-\beta} dE$ with $\beta = 2.2$, $E_1 = 1$ MeV, $E_c = 1$ MeV and $z_{\text{inj}} = 19$, evaluated with engine A gives $I_{1\text{ MeV}} = 8.1 \times 10^3$ against their quoted 1.5×10^4 — *a factor 1.9*. The no-inverse-Compton limit, $W = I_H = 36$ eV, gives 2.8×10^4 . Their value lies between the two and closer to the no-loss limit, consistent with finding H.

I — cross-consistency with the rest of the project.

quantity	here	elsewhere	diff.	source of the comparison value
$\langle n_H \rangle_{\text{com}} [\text{cm}^{-3}]$	1.8946×10^{-7}	1.8946×10^{-7}	0.00%	<code>crosscheck_reionization.py</code>
n_{He}/n_H	0.08130	0.08130	0.00%	<code>ionizing_rates_vs_...py</code>
$\sigma_{\text{H I}}(1 \text{ keV})$	1.141×10^{-23}	1.141×10^{-23}	0.0%	Karzas & Latter, <code>redo_common.py</code>
$\sigma_{\text{H I}}(1 \text{ keV})$	1.141×10^{-23}	2.208×10^{-23}	−48%	O&F fit, <code>ionizing_rates_vs_...py</code>
$W(1 \text{ keV}), x_e=10^{-4}$	35.29 eV	35.28 eV	+0.03%	<code>RESULTS.txt</code> stage 5
$W(1 \text{ keV}), x_e=10^{-1}$	89.76 eV	89.73 eV	+0.03%	<code>RESULTS.txt</code> stage 5

Audit result: the Verner et al. (1996) cross-sections selected for this work agree with the Karzas & Latter form already used in `Notebooks/redo` to 0.0 per cent at 1 keV, and disagree with the Osterbrock & Ferland near-threshold fit used in `ionizing_rates_vs_cosmic_rays.py` by a factor 1.94 at 1 keV and 5.0 at 10 keV. That script integrates the O&F fit to $10^4 \times$ threshold, far outside its range, while its parent document (`Reionization_State_of_the_Art.tex`, Eq. eq:fheat) states the cross-sections are Verner et al. (1996). *The text and the code disagree, and the code is the one that is wrong.*

Audit of the summary document itself. `verify_claims.py` was repaired (it opened `keymap.json`; the file is `verify_keymap.json`) and the extracted-text corpus `txt/` regenerated from the 60 PDFs. With that done, **117 numeric claims checked, 117 fully verified, 0 needing review** against the text of the cited papers. `crosscheck_reionization.py` also runs clean, both `sympy` dimensional checks returning `True`. Three further discrepancies between the summary’s conventions and `Notebooks/redo` were found and are listed in Sect. 5.

4 Final result

4.1 Per primary particle

Table 1: Ionizations per primary particle, engine A, $z = 10$, $x_e = 10^{-4}$. Numerical uncertainty is the engine A/B spread of Sect. 3F: 5 per cent over 300 eV–3 keV for $x_e \leq 10^{-2}$, degrading to 30 per cent at $x_e = 10^{-1}$; a factor $\lesssim 3$ over 1 MeV–1 TeV, $\lesssim 6$ at $x_e = 10^{-1}$. An entry of “—” is below the species threshold. Helium fully neutral (**flagged assumption**).

E [eV]	HYDROGEN		HELIUM	
	photon	electron	photon	electron
2×10^1	1	0.1166	—	0
10^2	2.206	2.404	0.6673	0.0709
10^3	26.29	26.56	2.443	1.781
10^4	255.5	255.7	21.25	20.58
10^5	2040	2040	172.3	171.6
10^6	6083	6082	520.7	519.9
10^8	9193	9193	798.2	797.5
10^{10}	1.044×10^4	1.044×10^4	910.5	909.8
10^{12}	1.096×10^4	1.096×10^4	955.4	954.7

Neutral helium absorbs most of the ionizing photons above 25 eV. This is the single most striking feature of the helium branch. Although helium is only 8.13 per cent of hydrogen by number, the ratio $\sigma_{\text{HeI}}/\sigma_{\text{HI}}$ from Eq. (2) reaches ~ 20 above 100 eV, so the branching ratio P_{HeI} of Eq. (7) is

E [eV]	20	24.7	30	50	100	300	10^3	5×10^4
$P(\text{HI})$	1.000	0.671	0.617	0.491	0.377	0.316	0.295	0.278
$P(\text{HeI})$	0.000	0.329	0.383	0.509	0.623	0.684	0.705	0.722

so between 50 eV and 50 keV *the majority of ionizing photons are absorbed by helium, not hydrogen*, under the fully-neutral-helium assumption. The helium ionizing emissivity is correspondingly 14 per cent of the hydrogen one at $z = 6$, not the 8.1 per cent the abundance ratio alone would suggest.

4.2 The emissivity comparison

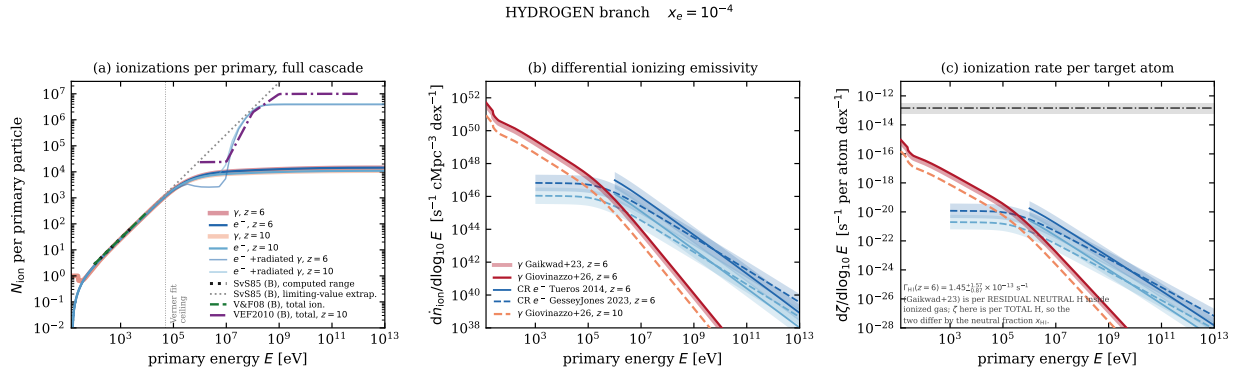


Figure 1: Hydrogen branch at $x_e = 10^{-4}$. (a) ionizations per primary, engine A solid, engine B in its published ranges; (b) differential ionizing emissivity; (c) ionization rate per target atom. The measured Γ_{HI} is shown for scale, with the warning that it is a rate per *residual neutral* hydrogen atom inside ionized gas whereas ζ here is per total hydrogen, the two differing by the neutral fraction. The equivalent figures for the other three x_e values, and the eight helium panels, are in `figures_emissivity/`.

Table 2: Integrated ionizing emissivity [ionizations $\text{s}^{-1} \text{cMpc}^{-3}$] and the cosmic-ray-electron share. Photon normalisation: [Giovinazzo et al. \(2026\)](#). The cosmic-ray range spans both injection spectra and $\eta_{\text{cr}} = 0.1\text{--}0.5$, $f_e = 0.01\text{--}0.02$.

z	x_e	HYDROGEN		HELIUM	
		$\dot{n}_{\text{ion}}^{\gamma}$	CR e^-/γ	$\dot{n}_{\text{ion}}^{\gamma}$	CR e^-/γ
6	10^{-4}	1.14×10^{51}	7.5×10^{-5}	1.62×10^{50}	4.4×10^{-5}
6	10^{-1}	1.00×10^{51}	4.6×10^{-5}	1.58×10^{50}	2.4×10^{-5}
8	10^{-4}	4.42×10^{50}	7.2×10^{-5}	—	—
10	10^{-4}	1.89×10^{50}	6.8×10^{-5}	—	—
12	10^{-4}	7.68×10^{49}	6.5×10^{-5}	—	—
15	10^{-4}	2.02×10^{49}	6.1×10^{-5}	—	—
15	10^{-1}	1.78×10^{49}	3.7×10^{-5}	—	—

The answer.

$$\frac{\dot{n}_{\text{ion}}^{\text{CR } e^-}}{\dot{n}_{\text{ion}}^{\gamma}} = (3.8\text{--}7.5) \times 10^{-5} \quad (\text{H}), \quad (2.0\text{--}4.4) \times 10^{-5} \quad (\text{He}), \quad (11)$$

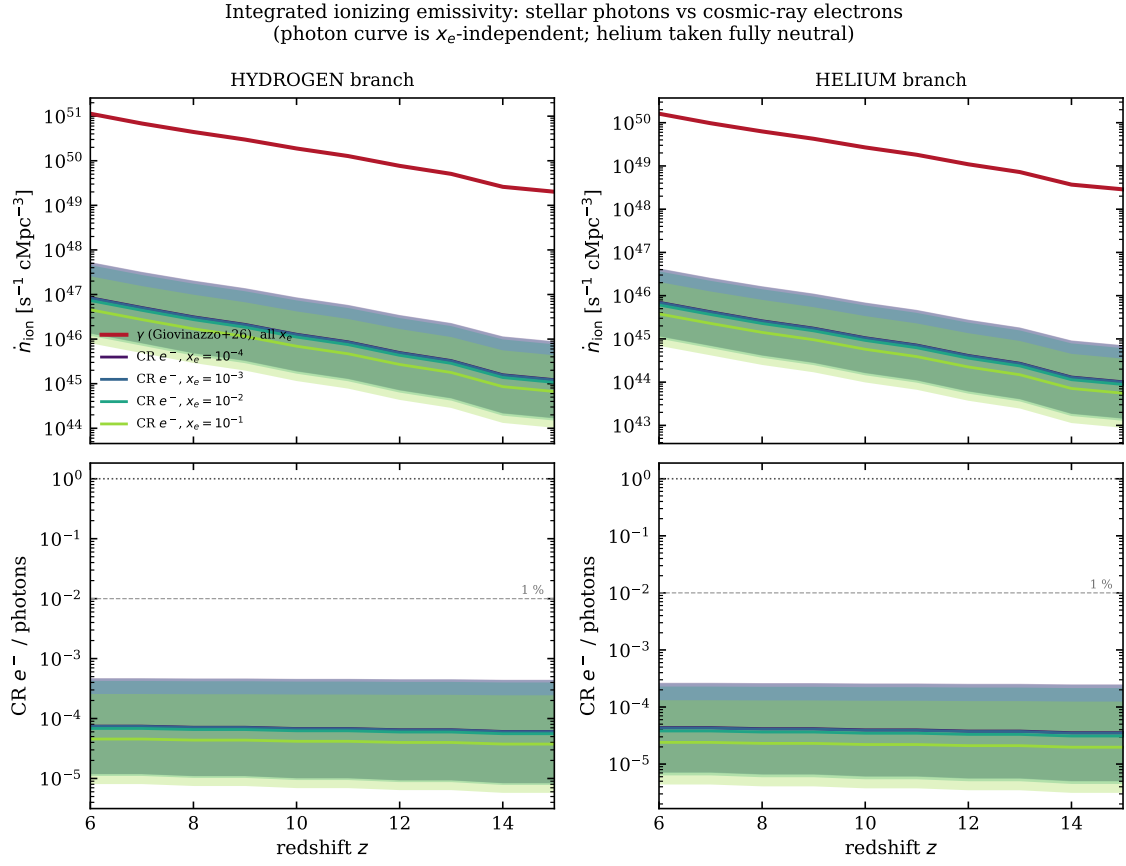


Figure 2: Integrated ionizing emissivity of the two populations and their ratio, for both branches and all four x_e . The redshift range is that tabulated by [Giovannozzo et al. \(2026\)](#); nothing is extrapolated beyond it.

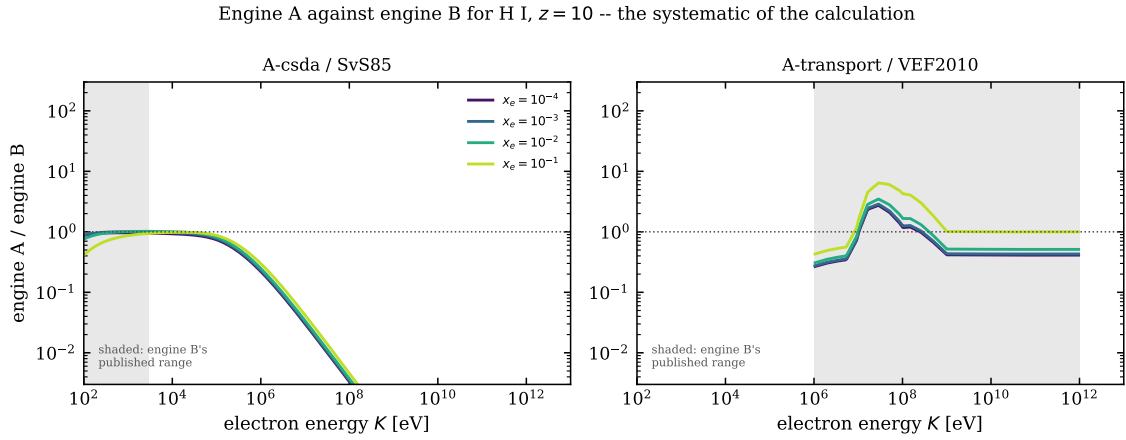


Figure 3: The systematic of the calculation: engine A against engine B over the whole energy axis, at $z = 10$, for H I. Shaded regions mark where engine B is inside its published range.

over $6 \leq z \leq 15$ and $10^{-4} \leq x_e \leq 10^{-1}$ using the geometric mean of the two injection prescriptions. Including their full span and the η_{cr}, f_e ranges, the hydrogen share is 5.8×10^{-6} – 4.7×10^{-4} and the helium share 3.1×10^{-6} – 2.7×10^{-4} . The ratio is remarkably flat: it falls by only 20 per cent from $z = 6$ to $z = 15$ (because both populations are tied to the same star formation history) and by a further 40 per cent from $x_e = 10^{-4}$ to 10^{-1} (because the cosmic-ray electrons lose an increasing share of their energy to Coulomb heating while the photons do not).

What the headline ratio does and does not depend on. Because the cosmic-ray budget is obtained by inverting $\dot{n}_{\text{ion}} = f_{\text{esc}} \xi_{\text{ion}} \rho_{\text{SFR}}$, the photon emissivity appears in *both* the numerator and the denominator of the ratio and cancels exactly. Swapping the [Giovinazzo et al. \(2026\)](#) normalisation for the [Gaikwad et al. \(2023\)](#) one changes $\dot{n}_{\text{ion}}$ by a factor 1.601 and changes the ratio by a factor 1.0000000000 (verified numerically). *The ± 0.09 dex uncertainty on the JWST emissivity, and Gaikwad’s asymmetric errors, therefore do not propagate into the answer at all.* What does propagate is

$$\frac{\dot{n}_{\text{ion}}^{\text{CR}e^-}}{\dot{n}_{\text{ion}}^{\gamma}} \propto \frac{\eta_{\text{cr}} f_e E_{\text{SN}}}{f_{\text{esc}} \xi_{\text{ion}} (M_*/N_{\text{CCSN}})}, \quad (12)$$

so the escape fraction is the dominant lever: adopting [Muñoz et al.](#)’s best-fit $\langle f_{\text{esc}} \rangle_{\text{ion}} = 0.05$ instead of the [Robertson et al. \(2015\)](#) fiducial 0.2 raises the ratio by a factor 4, to $\sim 1.5 \times 10^{-4}$ — still four orders of magnitude below the photon channel, but the flatness of the curves in Fig. 2 should not be read as insensitivity to f_{esc} . **f_e , the electron share of the cosmic-ray energy, is the least constrained link and has not been calibrated at high redshift.**

Where the two populations do compete. Panels (b) and (c) show the one place cosmic rays are not negligible: above $\sim 10^6$ eV the photon emissivity has fallen off the end of the $E^{-\alpha_s-1}$ spectrum, while the cosmic-ray spectrum is still there. The crossover is at $E \simeq 3\text{--}6 \times 10^6$ eV, and above it cosmic-ray electrons dominate the *differential* emissivity by orders of magnitude. This is energetically irrelevant — the integral is dominated by the decade above 13.6 eV — but it is the correct statement of where a cosmic-ray signature would have to be looked for.

5 Caveats and flagged items

1. **Helium fully neutral** at all x_e , by instruction. Upper bound; the consistent alternative would lower the helium emissivity by ≤ 10 per cent.
2. **Engine B’s helium column** inherits [Shull & van Steenberg](#)’s $n(\text{He})/n(\text{H}) = 0.1$ and their $x_{\text{He}^+} = x_{\text{H}^+}$, neither of which matches the assumptions used for engine A. The helium engine-A/B ratio is therefore quoted only as an order-of-magnitude consistency check.
3. **Verner et al. above 50 keV** is an extrapolation of a fit, using the correct non-relativistic asymptote. Marked on every panel (a).
4. **No spectral index from Giovinazzo et al. (2026)**; $\alpha_s = 2.0 \pm 0.6$ from [Gaikwad et al. \(2023\)](#) is used for both photon normalisations. Varying α_s over its 1σ range changes the H I emissivity integral by $\mp$ a few per cent and the He I one by more, since helium samples the spectrum above 24.6 eV.
5. **A-transport exists only at $x_e = 10^{-4}$** ; the other three x_e values carry A-csda alone above 1 MeV, which omits the re-absorbed radiated photons and is therefore a lower bound there.
6. **Three project-internal inconsistencies** surfaced by the audit and not repaired here, because they live in files outside this task’s scope: `Notebooks/redo/stage9_budget.py` uses $X = 0.760$ and m_p rather than $X_p = 1 - Y_p = 0.7546$ and m_{H} (a +0.99 per cent error in n_{H}); `igm_losses.N_HI_NORM` = $1.8 \times 10^3 \text{ m}^{-3}$ is +2.6 per cent above the Planck-consistent

1754.6 m⁻³; and `igm_losses.THRESHOLD_EV_ION` uses the Rydberg 13.6057 eV rather than the NIST hydrogen ionization potential 13.598 eV (+0.057 per cent).

7. **Two literature findings**, Sect. 3G and H, are reported as observations about published material, not as corrections.
8. **Not covered by any pass/fail test**, and reported rather than asserted: the H I/He I split (engine A gives $f_{\text{HeI}}/f_{\text{HI}} \rightarrow 0.0871$ against 0.0637 from [Shull & van Steenberg](#) rescaled to the Planck helium abundance, a 37 per cent difference that cannot be resolved because their helium fits assume $x_{\text{HeII}} = x_{\text{HII}}$); and the two photon emissivity normalisations, which are inputs that nothing here can falsify. The split is however shown to be insensitive to the shape of Λ : tripling the helium term moves f_{HeI} by 2.8 per cent at 100 eV and < 0.2 per cent above 1 keV, which is the derivation’s own claim.
9. **Provenance of the anchors**. Neither photon normalisation is a raw measurement: [Gaikwad et al. \(2023\)](#) calibrate against simulations of the Ly α forest, and [Giovinazzo et al. \(2026\)](#) infer $\dot{n}_{\text{ion}}$ from SED fitting with an assumed escape prescription.

Reproducibility

All code in `State of the art - Reionization/ionizing_emissivity/`. No stochastic method is used anywhere, so no random seed is involved.

- `emis_common.py` — constants, cosmology, Verner cross-sections.
- `emis_engine_A.py` and `emis_engine_A_xe.py` — engine A and its extended CSDA cache, `emis_engineA_cache_ext.npz`.
- `emis_engine_B.py` — SvS85, V&F08 and the parser for the VEF2010 appendix tables.
- `emis_sources.py` — the four normalisations and the IMF integral.
- `emis_figure.py` and `emis_summary_figure.py` → `figures_emissivity/` (8 branch figures + 2 summary figures).
- `emis_verify.py` → `emis_verify.log`; `emis_results.txt` holds the tabulated results.
- Papers added to `Agents/`: [Verner et al. \(1996\)](#), [Shull & van Steenberg \(1985\)](#), [Valdés & Ferrara \(2008\)](#), [Valdés et al. \(2010\)](#).

References

- Berezinskii, V. S., et al. 1990, *Astrophysics of Cosmic Rays* (Amsterdam: North-Holland).
- Blumenthal, G. R., & Gould, R. J. 1970, *Rev. Mod. Phys.*, 42, 237.
- Caprioli, D., & Spitkovsky, A. 2014, *ApJ*, 783, 91.
- Chabrier, G. 2003, *PASP*, 115, 763.
- Drury, L. O’C. 1983, *Rep. Prog. Phys.*, 46, 973.
- Drury, L. O’C., Markiewicz, W. J., & Völk, H. J. 1989, *A&A*, 225, 179.
- Fixsen, D. J. 2009, *ApJ*, 707, 916.
- Gaikwad, P., et al. 2023, *MNRAS*, 525, 4093 (arXiv:2304.02038).
- Gessey-Jones, T., et al. 2023, *MNRAS*, 526, 4262 (arXiv:2304.07201).

- Giovinazzo, E., et al. 2026, arXiv:2607.22834.
- Gould, R. J. 1972, *Physica*, 60, 145.
- Kim, Y.-K., & Rudd, M. E. 1994, *Phys. Rev. A*, 50, 3954.
- Kim, Y.-K., Santos, J. P., & Parente, F. 2000, *Phys. Rev. A*, 62, 052710.
- Muñoz, J. B., Mirocha, J., Chisholm, J., Furlanetto, S. R., & Mason, C. 2024, *MNRAS*, 535, L37 (arXiv:2404.07250).
- Planck Collaboration 2020, *A&A*, 641, A6 (arXiv:1807.06209).
- Robertson, B. E., et al. 2015, *ApJ*, 802, L19 (arXiv:1502.02024).
- Shull, J. M., & van Steenberg, M. E. 1985, *ApJ*, 298, 268.
- Stone, P. M., & Kim, Y.-K. 2002, *J. Phys. B*, 35, 1675.
- Tueros, M., del Valle, M. V., & Romero, G. E. 2014, *A&A*, 570, L4 (arXiv:1409.6225).
- Valdés, M., & Ferrara, A. 2008, *MNRAS*, 387, L8 (arXiv:0803.0370).
- Valdés, M., Evoli, C., & Ferrara, A. 2010, *MNRAS*, 404, 1569 (arXiv:0911.1125).
- Verner, D. A., Ferland, G. J., Korista, K. T., & Yakovlev, D. G. 1996, *ApJ*, 465, 487 (arXiv:astro-ph/9601009).
- Woosley, S. E., & Weaver, T. A. 1995, *ApJS*, 101, 181.